

An ABP estimate for nonlocal equations

Abstract

We prove an L^p maximum estimate for uniformly elliptic nonlocal equations with even kernels. The kernel coefficients are represented by an integral of nonnegative spatial densities against bounded jump profiles, subject to a uniform moment bound. Integrable corrections are permitted, and the constants are uniform as the order approaches two.

1 The estimate

For $n \geq 1$ and $0 < \sigma < 2$, write

$$\nu_\sigma(dz) = (2 - \sigma)|z|^{-n-\sigma} dz, \quad \delta_z u(x) = u(x+z) + u(x-z) - 2u(x), \quad Lu(x) = \frac{1}{2} \int \delta_z u(x) a(x, z) \nu_\sigma(dz).$$

Let $w : [0, \infty) \rightarrow [0, \infty)$ be increasing and convex, with $w(0) = 0$, $w(t) > 0$ for $t > 0$, and

$$w(2t) \leq d w(t) \quad (t \geq 1), \quad \ell := \int_1^\infty \frac{dt}{w(t)} < \infty. \quad (1)$$

Let (Ω, μ) be a standard probability space, $0 < \lambda \leq \Lambda_0 < \infty$, and $\lambda \leq \Lambda < \infty$. We assume

$$\begin{aligned} a &= a_0 + b, \quad a_0(x, z) = \int_\Omega \rho(x, t) \phi(z, t) \mu(dt), \\ \rho &\geq 0, \quad |\phi| \leq 1, \quad \phi(-z, t) = \phi(z, t), \quad b(x, -z) = b(x, z), \\ E &:= \sup_x \int_\Omega w(1 + \rho(x, t)) \mu(dt) < \infty, \quad K := \sup_x \int |b(x, z)| \nu_\sigma(dz) < \infty, \\ \lambda &\leq a_0(x, z) \leq \Lambda_0, \quad \lambda \leq a(x, z) \leq \Lambda. \end{aligned} \quad (2)$$

All coefficient maps are jointly Borel measurable. The bounds hold at every spatial point and almost every jump. The parameters E and K are part of the hypotheses.

Fix $\sigma_0 \in (0, 2)$ and set

$$\varepsilon = \min \left\{ \frac{1}{4}, \frac{\sigma_0}{2n}, \frac{\lambda^2}{32(d+1)E^2\ell^2} \right\}, \quad q = 1 + \varepsilon, \quad p = \frac{q}{q-1}. \quad (3)$$

Theorem 1. *Let $R > 0$, $\sigma \in [\sigma_0, 2)$, and assume (1)–(2). If $u \in C^2(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$, $f \in L^p(B_R)$, and*

$$-Lu \leq f \quad \text{a.e. in } B_R, \quad u \leq 0 \quad \text{on } \mathbb{R}^n \setminus B_R,$$

then

$$\|u_+\|_{L^\infty(B_R)} \leq C \left(1 + \frac{C_n R^\sigma K}{\lambda} \right) R^{\sigma-n/p} \|f_+ \mathbf{1}_{\{u>0\}}\|_{L^p(B_R)}. \quad (4)$$

Here C depends only on $n, \sigma_0, \lambda, E, d, \ell$. In particular, p and C are independent of (Ω, μ) and uniform for $\sigma \in [\sigma_0, 2)$.

2 Proof

Put $D = E\ell^2$ and $S = (d+1)E$. Define the coefficient entropy by

$$j(a) = \left(\int_{1+a}^{\infty} \frac{dt}{w(t)} \right)^{-1}, \quad J(a) = \int_0^a j(t) dt, \quad P(A) = E + \int_{\Omega} J(A) d\mu.$$

Since $j(0) = \ell^{-1}$,

$$J''(a) = \frac{j(a)^2}{w(1+a)}, \quad \frac{J'(a)^2}{J''(a)} = w(1+a), \quad \frac{1}{J''(a)} \leq \ell^2 w(1+a).$$

Also $\int_r^{2r} dt/w(t) \geq r/(dw(r))$, so $j(a) \leq dw(1+a)/(1+a)$. The ratio $w(r)/r$ is nondecreasing, giving $J(a) \leq dw(1+a)$. Thus, whenever $\int w(1+A) d\mu \leq E$,

$$\begin{aligned} E \leq P(A) \leq S, \quad |DP(A)[H]|^2 &\leq P(A) \mathcal{Q}_A(H), \\ \left| \int H\phi(z, \cdot) d\mu \right|^2 &\leq D \mathcal{Q}_A(H), \quad \mathcal{Q}_A(H) = \int J''(A) H^2 d\mu. \end{aligned} \quad (5)$$

The last two inequalities are weighted Cauchy-Schwarz.

For $r > 0$ and $B \geq 0$, set $F(r, B) = r^q P(B/r)$. Along an affine segment $Z_t = (r_t, B_t)$, write $A_t = B_t/r_t$. Since $\ddot{A}_t = -2\dot{r}_t \dot{A}_t/r_t$,

$$\begin{aligned} \frac{d^2}{dt^2} F(Z_t) &= q(q-1)r_t^{q-2}\dot{r}_t^2 P(A_t) + 2(q-1)r_t^{q-1}\dot{r}_t DP(A_t)[\dot{A}_t] + r_t^q \mathcal{Q}_{A_t}(\dot{A}_t) \\ &\geq q(q-1)(a-b/q)^2 + b^2/q, \end{aligned}$$

where $a = r_t^{(q-2)/2} |\dot{r}_t| \sqrt{P(A_t)}$ and $b = r_t^{q/2} \sqrt{\mathcal{Q}_{A_t}(\dot{A}_t)}$. Consequently

$$\mathcal{D}_F(Z_0, Z_1) := \langle DF(Z_1) - DF(Z_0), Z_1 - Z_0 \rangle \geq \frac{1}{q} \int_0^1 r_t^q \mathcal{Q}_{A_t}(\dot{A}_t) dt. \quad (6)$$

Only segments within the coefficient constraints are used; these constraints are convex.

Choose a smooth $0 \leq \chi \leq 1$, zero on $[0, 1]$ and one on $[2, \infty)$, and define

$$\eta(0) = \eta'(0) = 0, \quad \eta''(r) = q(q-1)r^{q-2}\chi(r).$$

Then $\eta = 0$ below 1, $\eta'(r) \leq q(r^{q-1} - 1)_+$, $r^2\eta''(r) \leq q(q-1)r^q$, and $r^q \leq \eta(r) + 2qr$. If $B_i = r_i A_i$ and $\int A_i \phi(z, \cdot) d\mu \geq \lambda_*$, interpolation, Young's inequality, and (5)–(6) give

$$\begin{aligned} &\left(\int (B_1 - B_0) \phi(z, \cdot) d\mu \right) (\eta'(r_1) - \eta'(r_0)) \\ &\geq \frac{\lambda_*}{2} \mathcal{D}_\eta - \frac{q^2(q-1)D}{2\lambda_*} \mathcal{D}_F, \end{aligned} \quad (7)$$

where $\mathcal{D}_\eta = (r_1 - r_0)(\eta'(r_1) - \eta'(r_0))$. Indeed, the interpolation error costs at most

$$\frac{q(q-1)}{2\lambda_*} \int_0^1 r_t^q \left| \int \dot{A}_t \phi d\mu \right|^2 dt.$$

Lemma 1 (Bounded-density adjoint estimate). *Suppose (2) holds with $0 \leq \rho \leq T$ and lower bounds $\lambda_* := \lambda/2$. If $v \geq 0$ is a finite measure supported in $\overline{B_1}$, $G \geq 0$ is a finite measure, and $-L^*v \leq G$ globally, then*

$$\|v\|_{L^q(\mathbb{R}^n)} \leq C(1 + C_n K/\lambda) G(\mathbb{R}^n).$$

The constant is independent of T .

Proof. First take $b = 0$. Choose $0 \leq \psi \leq (4 - |x|^2)_+$, smooth and compactly supported, equal to $4 - |x|^2$ on $B_{3/2}$. For $x \in \overline{B_1}$, $\delta_z \psi \leq 0$ for all z , and $\delta_z \psi = -2|z|^2$ for $|z| < 1/2$. Hence $-L\psi \geq c_n \lambda_*$ and testing gives

$$v(\mathbb{R}^n) \leq C_n G(\mathbb{R}^n) / \lambda_*. \quad (8)$$

Normalize $G(\mathbb{R}^n) = 1$; the zero-mass case follows from (8).

Evolve v , $B = \rho v$, and G by $\partial_h W_h = \mathcal{H}W_h$, where $\mathcal{H}W = \frac{1}{2} \int \delta_z W \nu_\sigma(dz)$. Its positive convolution kernel preserves mass and satisfies

$$\|W_h\|_\infty \leq C(n, \sigma_0) h^{-n/\sigma} \|W\|_{\mathcal{M}}.$$

Indeed, the Fourier symbol is $-d_{n,\sigma} |\xi|^\sigma$, with $d_{n,\sigma}$ uniformly positive and bounded; Fourier inversion bounds the kernel by $Ch^{-n/\sigma}$. Set $A_h = B_h/v_h$ when $v \neq 0$. It is an average of ρ , so all coefficient constraints persist. The adjoint expression

$$L_0^* v = \mathcal{T}B, \quad \mathcal{T}B = \frac{1}{2} \int \int_\Omega \delta_z B(\cdot, t) \phi(z, t) \mu(dt) \nu_\sigma(dz)$$

commutes with the heat flow. Thus $-\mathcal{T}B_h \leq G_h$.

Put $I_1(h) = \int \eta(v_h) dx$ and $I_2(h) = q(q-1) \int F(v_h, B_h) dx$. Differentiation and symmetrization give

$$-I_1' = \frac{1}{2} \iint \mathcal{D}_\eta dx \nu_\sigma(dz), \quad -I_2' = \frac{q(q-1)}{2} \iint \mathcal{D}_F dx \nu_\sigma(dz),$$

with states at $x, x+z$. Testing the evolved inequality with $\eta'(v_h)$ and using (7) yields

$$-I_1' \leq \frac{qD}{\lambda_*^2} (-I_2') + \frac{2}{\lambda_*} \int \eta'(v_h) G_h dx. \quad (9)$$

By (8) and heat dispersion, $\|v_h\|_\infty \leq Ch^{-n/\sigma}$. Hence $I_1(h) = 0$ for large h , and

$$\int_0^\infty \int \eta'(v_h) G_h dx dh \leq \int_0^\infty q((Ch^{-n/\sigma})^{q-1} - 1)_+ dh \leq C,$$

since $n(q-1)/\sigma \leq 1/2$. Integrating (9) from $\delta > 0$ and using (5) gives

$$I_1(\delta) \leq \frac{q^2(q-1)DS}{\lambda_*^2} \int v_\delta^q dx + C \leq \frac{25}{128} I_1(\delta) + C.$$

Absorption and weak L^q compactness show that v has the claimed density.

Here B is a finite Bochner $L^2(\mu)$ -valued measure. At fixed T , extend J outside $[0, T]$ with a Lipschitz derivative; on the cone $0 \leq B \leq Tr$, set $F(0, 0) = 0$ and note $\|DF(r, B)\| \leq C_T r^{q-1}$. First truncate the jumps, symmetrize, and then use convergence of the generators in L^1 and nonnegativity of the entropy dissipation. Heat-smoothed finite measures have integrable spatial derivatives, so these passages and differentiation are valid. Compact spatial cutoffs justify convolving the distributional inequality. These regularizations introduce no quantitative constant depending on T .

For general b , let $J_b = L - L_0$. Its adjoint applied to v is the signed measure

$$\gamma(E) = \int v(dx) \int (\mathbf{1}_E(x+z) - \mathbf{1}_E(x)) b(x, z) \nu_\sigma(dz).$$

It has total mass zero and variation at most $2Kv(\mathbb{R}^n)$; thus $\gamma_+(\mathbb{R}^n) \leq Kv(\mathbb{R}^n)$. Since $-L_0^* v \leq G + \gamma_+$, apply the principal-part estimate and the mass bound for the full operator. $\square$

Proof of Theorem 1. First let $R = 1$, $\rho \leq T$, and mollify coefficients in x . Fix $x_0 \in B_1$. Truncate jumps to $|z| > r$ and set

$$k_r(x) = \int_{|z|>r} a(x, z) \nu_\sigma(dz), \quad P_r F(x) = k_r(x)^{-1} \int_{\substack{|z|>r \\ x+z \in B_1}} F(x+z) a(x, z) \nu_\sigma(dz).$$

Jumps with $|z| > \max(2, r)$ leave the ball, so $\|P_r\|_\infty < 1$. The positive inverse of $-L_r$ with zero exterior values is $S_r F = \sum_{j \geq 0} P_r^j(F/k_r)$. Represent $F \mapsto S_r F(x_0)$ on $C(\overline{B_1})$ by a positive measure v_r . For a nonnegative global test ψ , exterior positivity gives $S_r(-L_r \psi) \leq \psi$, hence

$$\int (-L_r \psi) dv_r \leq \psi(x_0).$$

The quadratic barrier used in (8) bounds $v_r(\mathbb{R}^n)$ uniformly as $r \downarrow 0$ at fixed σ . Taking a weak limit and using uniform convergence $L_r \psi \rightarrow L \psi$ gives $-L^* v \leq \delta_{x_0}$. Lemma 1 bounds $\|v\|_q$.

For a smooth convex approximation ζ of the positive part, zero on $(-\infty, 0]$ and with $0 \leq \zeta' \leq 1$, the function $u_\zeta = \zeta(u)$ vanishes outside B_1 . Thus $S_r(-L_r u_\zeta) = u_\zeta$, and passage to the limit gives

$$u_\zeta(x_0) = \int (-L u_\zeta) dv \leq \|v\|_q \|(-L u)_+ \mathbf{1}_{\{u > 0\}}\|_p.$$

Here $L u_\zeta \geq \zeta'(u) L u$, and $L u_\zeta \geq 0$ wherever $u \leq 0$. Let $\zeta(u) \rightarrow u_+$. Spatial mollification preserves the moment, rate and ellipticity bounds, and $L_\epsilon u \rightarrow L u$ in $L^p(B_1)$ by dominated convergence, so the estimate holds for measurable bounded densities.

For unbounded ρ , use $\rho_T = \min(\rho, T)$. Convexity of w and (1) imply $w(t)/t \rightarrow \infty$, and therefore

$$\sup_{x,z} |a_{0,T} - a_0| \leq \sup_x \int \rho \mathbf{1}_{\{\rho > T\}} d\mu \leq E \frac{1+T}{w(1+T)} =: d_T \rightarrow 0.$$

For $d_T \leq \lambda/2$, both truncated kernels retain the lower bound λ_* . Lemma 1 and the preceding proof are uniform in T . For the fixed bounded C^2 function u ,

$$\sup_{B_1} |L_T u - L u| \leq \frac{d_T}{2} \sup_{B_1} \int |\delta_z u| \nu_\sigma(dz) \rightarrow 0.$$

Apply the truncated estimate with right-hand side $f + o(1)$ and pass to the limit. Finally, $x = R y$ preserves the representation and moment bounds, multiplies the rate by R^σ , and gives the factor $R^{\sigma-n/p}$ in (4). $\square$

3 Examples of kernels

All maps in the examples below are Borel measurable. Ellipticity means the two positive bounds in (2).

Moment weights

For every $\alpha > 0$, the weight $w(t) = t[\log(e+t)]^{1+\alpha}$ satisfies (1). Thus a uniform $L \log^{1+\alpha} L$ moment of ρ suffices. For $\beta > 0$, the convex doubling weight

$$w_\beta(t) = \int_0^t \log(e+s) [\log \log(e+s)]^{1+\beta} ds$$

is comparable at infinity to $t \log t (\log \log t)^{1+\beta}$ and also satisfies (1).

Finite profiles

Every even translation-invariant kernel $\lambda \leq a_0(z) \leq \Lambda_0$ has the representation (2): take a one-point probability space, $\rho = \Lambda_0$ and $\phi = a_0/\Lambda_0$.

More generally, suppose

$$a_0(x, z) = \sum_{j=1}^N c_j(x) k_j(z), \quad |c_j| \leq M, \quad |k_j| \leq 1, \quad k_j(-z) = k_j(z),$$

and a_0 is elliptic. On $\Omega = \{1, \dots, N\} \times \{-1, 1\}$ with uniform probability, set

$$\rho(x, j, \tau) = 2N(\tau c_j(x))_+, \quad \phi(z, j, \tau) = \tau k_j(z).$$

Then $a_0 = \int \rho \phi d\mu$ and $E \leq w(1 + 2NM)$. This includes kernels constant on a finite symmetric Borel partition $D_1, \dots, D_N$ of $\mathbb{R}^n \setminus \{0\}$,

$$a_0(x, z) = a_j(x) \quad (z \in D_j), \quad \lambda \leq a_j(x) \leq \Lambda_0,$$

and quadratic angular kernels

$$a_0(x, z) = \frac{z^T Q(x) z}{|z|^2}, \quad Q(x) = Q(x)^T, \quad \lambda I \leq Q(x) \leq \Lambda_0 I.$$

For the latter, use the n^2 profiles $z_i z_j / |z|^2$. The jump cells may depend on both direction and length; no regularity of their boundaries is required.

Mixtures and integrable corrections

Let $k(t, z)$ be even in z , with $\lambda \leq k \leq \Lambda_0$, and let $r_x \geq 0$ satisfy $\int r_x d\mu = 1$. The mixture

$$a_0(x, z) = \int_{\Omega} r_x(t) k(t, z) \mu(dt)$$

satisfies (2) whenever $\sup_x \int w(1 + r_x) d\mu < \infty$: take $\rho = \Lambda_0 r_x$ and $\phi = k/\Lambda_0$, and use doubling to bound the rescaled moment. The profiles in the general representation may be signed and need not be individually elliptic.

The correction b may depend arbitrarily on x, z , subject to its rate bound and ellipticity. For example, suppose $a = a_0$ for $|z| < r_0$, while a, a_0 are even and lie between λ and Λ elsewhere. Then

$$K \leq 2\Lambda \frac{(2 - \sigma)|\mathbb{S}^{n-1}|}{\sigma} r_0^{-\sigma}.$$

Thus the theorem allows arbitrary measurable kernels on jumps of length at least r_0 .

The estimate also holds for $-\sup_j L_j u \leq f$ when the countable family has a common probability space and profiles, with uniform bounds in (2): select a measurable $j(x)$ within ϵ of the supremum, apply the theorem with $f + \epsilon$, and let $\epsilon \downarrow 0$.

Countably many profiles

Take $\Omega = \{0, 1, \dots\}$ with $\mu_0 = 1/2$ and $\mu_i = c(i+1)^{-2}$ for $i \geq 1$, choosing c so that $\sum_i \mu_i = 1$. Put $\beta_i = [\log(e/\mu_i)]^{-2}$. Choose disjoint Borel cells $E_0, E_1, \dots \subset B_1$ of positive measure. For each $N \geq 1$ and $\theta \in \{-1, 1\}^N$, choose a disjoint symmetric Borel jump cell $D_{N, \theta}$ of infinite ν_σ measure. Such cells are obtained by partitioning the dyadic annuli approaching zero into countably many infinite subfamilies.

Set

$$\rho(x, 0) = 4, \quad \rho(x, i) = \frac{\beta_i}{\mu_i} \mathbf{1}_{E_i}(x) \quad (i \geq 1).$$

Set $\phi(z, 0) = 1$ everywhere. On $D_{N, \theta}$ set $\phi(z, i) = \theta_i$ for $i \leq N$ and zero for $i > N$; outside these jump cells set all positive-index profiles to zero. Then

$$a(x, z) = 2 + \beta_i \theta_i \quad (x \in E_i, z \in D_{N, \theta}, i \leq N), \quad a = 2 \quad \text{otherwise.}$$

This kernel is even, $1 \leq a \leq 3$, and has $b = 0$. For $w(t) = t \log^2(e + t)$ its moment is uniform, since

$$\mu_i w(1 + \beta_i / \mu_i) \leq C(1 + \mu_i \log^2(e/\mu_i)).$$

For this representation, $\sup_x \int \rho^s d\mu = \infty$ for every $s > 1$, because $\mu_i^{1-s} \beta_i^s \rightarrow \infty$.

Ellipticity does not imply the representation

Define another kernel on the same cells by

$$\tilde{a}(x, z) = 2 + \theta_i \quad (x \in E_i, z \in D_{N,\theta}, i \leq N), \quad \tilde{a} = 2 \quad \text{otherwise.}$$

It is even and satisfies $1 \leq \tilde{a} \leq 3$, but admits no representation (2), even with a correction of finite rate.

To prove this, average a hypothetical representation over E_i , obtaining densities r_i and corrections b_i with uniform moment and rate bounds. Put

$$\epsilon_T = \sup_{i \geq 0} \int (r_i - \min(r_i, T)) d\mu \longrightarrow 0, \quad h_i = r_i - r_0.$$

The limit follows from $w(t)/t \rightarrow \infty$. For fixed N, θ and $0 < \delta < 1/2$, choose $z \in D_{N,\theta}$ with $|b_i(z)| \leq \delta$ for $0 \leq i \leq N$; their bad sets have finite jump measure. Subtracting the baseline row gives

$$\left\| \sum_{i \leq N} \theta_i h_i \right\|_{L^1(\mu)} \geq (1 - 2\delta)N.$$

Truncate r_i at T and average over the signs. The truncated sum has expected L^1 norm at most $T\sqrt{N}$, and the tails cost at most $2N\epsilon_T$. Hence

$$(1 - 2\delta)N \leq T\sqrt{N} + 2N\epsilon_T,$$

a contradiction as $N \rightarrow \infty$ and then $T \rightarrow \infty$. This example concerns the kernel hypotheses; it does not assert failure of a maximum estimate for $\tilde{a}$.